ELE 717

Diffusion and Ion Implantation.

In Semi-Conductor manufacturing, diffusion and ion Implantation are two major methods used to introduced Controlled Impurities (dopants) into a semi-conductor such as Silicon. The purpose is to change the electrical properties of the material and form n-type and p-type regions required for devices such as diodes, transistors MOSFETs and Integrated Circuits.

Diffusion Implantation Mechanism

Diffusion is the movement of dopant atoms from a region of high Concentration to a region of Low Concentration due to thermal energy. In Semi-Conductor processing, a Silicon wafer is heated to a high temperature and exposed to a dopant source. Dopant atoms enter the Silicon Surface and gradually move into the Crystal.

Common dopant include.

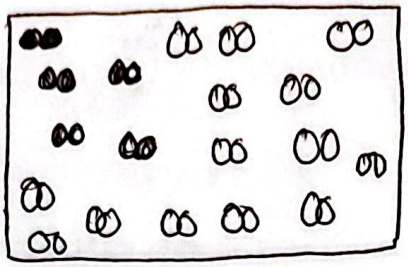
Semi-Conductor	Dopant	Type.
(1) Silicon	phosphorus(ph)	n type.
(2) Silicon.	Arsenic(As)	n type.
(3) "	Antimony(Sb)	"
(4) "	Boron (B)	p type
(5) "	Gallium (Ga)	p type.

Note.

Net diffusion: Random motion carries molecules both ways, but more move from the more Concentrated side to the less Concentrated side.

Dynamic equilibrium: The two sides have Similar Concentration, while molecules keep moving randomly in both directions.

Concentration difference: 21 of 32 Oxygen molecules begin on the Left; creating an Concentration gradient.



Basic principle: The diffusion process can be represented as:

- Dopant source → Silicon surface → Dopant penetration → Desired dopant profile.

- The dopant atoms move because of a concentration gradient.

Fick's first law: for a steady state diffusion process.

$$J = -D\frac{dc}{dx}$$

Where J = dopant flux

D = diffusion Co-efficients.

C - dopant Concentration.

x - distance into the semi-Conductor.

The negative sign indicates that diffusion occurs from high Concentration toward low Concentration.

Diffusion Co-efficient: The diffusion Co-efficient depent strongly on temperature.

$$D = D_0 e^{-E_a/KT}$$

where D_0 = diffusion Constant

E_a = activation energy

K = Boltzmann Constant

T = absolute temperature

∴ Increasing temperature greatly increase the diffusion rate.

Types of Diffusion -

There are type Important diffusion processes used in Semi-Conductor fabrication.

(A) Constant-Source diffusion: In Constant-Source diffusion, the Semi Conductor Surface is maintained at approximately Constant dopant Concentration.

The Concentration profile is:

$$C(x,t) = C_s \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

where

C_s = Surface Concentration

x = depth

D = Diffusion Coefficient

t = diffusion time

erfc = Complementary error function.

Characteristics -

1. Dopant Source remains available throughout the process.
2. Surface Concentration remains approximately Constant
3. produces an erfc Concentration profile.
4. often used during the Initial doping Steps.

(B) Limited-Source or drive-in diffusion: After a dopant has been introduced into the Surface, the wafer may be heated without continuously Supplying more dopant. The dopant Spreads deeper into the Semi-Conductor.

profile is approximately Gaussian.

$$C(x,t) = \frac{Q}{\sqrt{\pi Dt}} e^{-x^2/4Dt}$$

where

Q = total dopant dose per unit area

D = diffusion Co-efficient

t = time

Characteristics. ① Dopant penetrates deeper

② Surfaces Concentration decreases.

③ The total dopant quantity is approximately Conserved

④ Used to Obtain a desired Junction depth.

Step by step Diffusion process..

① Wafer preparation.

A high-purity Silicon wafer is cleaned to remove:

ⓐ dust ⓑ organic Contamination ⓒ metallic Impurities ⓓ native Surface Contamination

② Dopant Source preparation. It is suitable dopant source is Selected. Examples Boron - p type, phosphorus - n type, Arsenic - n type.

③ Heating The wafer is placed in a diffusion furnace and heated, typically to a high temperature..

④ Dopant Introduction: The dopant is supplied to the surface of the Silicon.

⑤ Dopant penetration: Thermal energy allows dopant atoms to move from the Surface into the Silicon lattice.

⑥ Drive-in Additional heating may be used to drive the dopants deeper into the wafer.

⑦ Cooling - The wafer is Cooled under Controlled Conditions.

(8) Characterization : The resulting dopant profile is measured to determine :

(1) Junction depth (2) Sheet resistance (3) dopant Concentration (4) electrical properties.

Diffusion Mechanism at the atomic level

Dopant atoms can move through the Silicon Crystal mainly by two mechanisms.

(A) Vacancy mechanism : A dopant atom moves into a vacant position in the Crystal Lattice :

Atom → Vacancy → New Lattice position.

(B) Interstitial mechanism An atom moves through spaces between the normal lattice positions.

Some dopants can move through interstitial sites or through interactions involving both interstitials and Vacancies. These determines how quickly different dopant diffuse

Advantages of Diffusion.

(1) Relatively simple process (11) Can process many wafers simultaneously (111) Suitable for high-temperature processing. (IV) Good for introducing large dopant concentration (V) Historically important in Semiconductor fabrication.

Disadvantages of Diffusion.

(1) Requires high temperature (11) Dopants spread laterally as well as vertically (111) less precise depth control (14) Difficult to form extremely shallow junctions (V) High-temperature exposure can affect previously fabricated structure.

Calculation

Q1 A phosphorus dopant is diffused into silicon. If the diffusion coefficient is $2\times10^{-13}\ cm^3/s$ and diffusion time is 500s Calculate the approximate diffusion distance using.

$$L = \sqrt{2Dt}$$

Data

$D = 2\times10^{-13}\ cm^3/s$.

$t = 500s$

$L = ?$

$$L = \sqrt{2\times(2\times10^{-13})\times(500)}$$

$$L = \sqrt{2\times10^{-10}}$$

$$L = 1.414\times10^{-5}\ cm$$

Convert to micrometre $1cm = 10^4\ \mu m$

$$= L = 1.414\times10^{-5}\times10^4$$

$$L = 0.141\ \mu m.$$

Q2 The concentration gradient of a dopant in Silicon is $4\times10^{18}\ cm^{-4}$ and diffusion co-efficient is $3\times10^{-14}\ cm^2/s$ Calculate the diffusion flux.

using fick's first Law.

$$J = -D\frac{dc}{dx}$$

Data

$D = 3\times10^{-14}$

$\frac{dc}{dx} = 4\times10^{18}\ cm^{-4}$.

$$J = -(3\times10^{-14})(4\times10^{18})$$

$$J = -12\times10^{-14+18}$$

$$= -12\times10^4$$

$$\therefore J = 1.2\times10^5\ atom/cm^2 s$$

Interpretation

The negative sign indicates that the dopant moves from the region of higher concentration to lower concentration.

Q4 Calculate Diffusion Coefficient at a Given Temperature

The diffusion Coefficient is given by $D = D_0 e^{-E_a/KT}$ for a dopant

Silicon :- $D_0 = 0.5 cm^2/s$, $E_a = 3.5 eV$ $T = 1273 K$.

$K = 8.617 \times 10^{-5} eV/K$

Calculate D ?

$$D = 0.5 e^{-\frac{3.5}{(8.617 \times 10^{-5})(1273)}}$$

$$KT = (8.617 \times 10^{-5}) \times (1273) = 0.1097 eV$$

$$D = 0.5 e^{-35/0.1097}$$

$$D = 0.5 e^{-31.90}$$

$$D = 6.9 \times 10^{-15} cm^2/s.$$

Q5 A Silicon wafer is subjected to Constant-Source diffusion The Surface Concentration is 1×10^{20} atom/cm^3 the diffusion Coefficient is $1 \times 10^{-13} cm^3/s$ time is 3600 s find the Concentration at a depth of $x = 0.2 \mu m$

Formula $= C(x,t) = C_s \, erfc\left(\frac{x}{2\sqrt{Dt}}\right)$

Note Table.

$erfc(0.527)$

$1 - 0.544$

0.456

Convert $0.2 \mu m = 2 \times 10^{-5} cm$

$$Dt = \sqrt{(1 \times 10^{-13}) \times 3600} = \sqrt{3.6 \times 10^{-10}} = 1.897 \times 10^{-5} cm.$$

$$\therefore \frac{x}{2\sqrt{Dt}} = \frac{2 \times 10^{-5}}{2(1.897 \times 10^{-5})} = 0.527$$

From an erfc table $= erfc(0.527) = 0.456$

$C = (1 \times 10^{20})(0.456) = 456 \times 10^{19}$ atom/cm^3

(7)

Q6 A dopant dose of 2×10^{15} atoms/cm^3 is introduced into Silicon, after drive-in diffusion 5×10^{14} cm^2/s and have a time of 10000s Calculate the Concentration at $x=0$ using.

$$C(x,t) = \frac{Q}{\sqrt{\pi \Delta t}} e^{-x^2/4\Delta t}$$

At $x=0$

$$e^{-0} = 1$$

$$\therefore \quad C(0,t) = \frac{Q}{\sqrt{\pi \Delta t}}$$

$$C(0,t) = \frac{2\times10^{15}}{\sqrt{(3.142)(5\times10^{-14})(10000)}}$$

$$\Delta t = 5\times10^{-11}$$

$$\pi \Delta t = 1.571\times10^{-10}$$

$$\sqrt{\pi \Delta t} = 1.253\times10^{-5}$$

$$\therefore \quad C(0,t) = \frac{2\times10^{15}}{1.253\times10^{-5}}$$

$$C(0,t) = 1.60\times10^{20} \text{ atoms/cm}^3$$

Note

Why $x=0$ x represents the distance from the Silicon Surface

$x=0$ → exactly at the Surface
$x=0.1\mu m$ = 0.1 below the Surface.
$0.5\mu m$ = 0.5 below the Surface
1 — below the Surface
$x>0$ mean below the Surface.

} it asked for the Surface Concentration we set as 0

$e^0 = 1$.

Ion Implantation Mechanism

Ion Implantation mechanism is a Semi Conductor doping techniques in which dopant atoms are Converted into ions, accelerated to high energy, and physically injected into a Semi-Conductor wafer. Unlike thermal diffusion, ion implantation allow the dopant dose and implantation depth to be Controlled very precisely.

Basic Ion Implantation Mechanism.

1. flowchart

Dopant Source → Ionizer → mass analyzer → Accelerator → Beam → wafer.

Main Component:

- Ion Source or Dopant Source.
- Mass analyzer
- Acceleration Systen
- Beam focusing System
- Scanning System
- Semiconductor wafer
- Vacuum Chambe

Step by Step Ion Implantation-

Step 1 dopant generation: A Dopant-Containing material is introduced into an ion Source e.g Boron, phosphorus, Arsenic. The atom are Converted into positively charged ions.

Step 2 Ion extraction-: Electric field extract the ions from the Ion Source such as Boron phosphorus, Arsenic etc.

Step 3 Mass analysis: The ion Beam passes through a magnetic electromagnetic mass analyzer. This Selects ions according to the mass to charge ratio. This is Important because it prevents unwanted species from being Implanted.

Step 4 Acceleration. The Selected ions are acceleration through a high electric potential. The ion kinetic energy is approximately:

$$E = qV$$

where
E = ion energy
q = ion charge
V = accelerating Voltage

For a singly charged ion:

$$E = eV$$

Step 5 : Beam focusing and Scanning-

The ion beam is focused and Scanned across the wafer so that the dopant is distributed uniformly.

Step 6 Implantation The energetic ions enter the Silicon Crystal As they travel through the Silicon, they lose energy through Collision with Silicon nuclei and Silicon electrons. Eventually, the ions Stop at a particular average depth.

Step 7 : Annealing- : Ion Implantation damages the Crystal lattice because energetic ion displace Silicon atoms. The wafer is therefore Subjected to thermal annealing: Repairs Crystal damage, activates dopant atoms, places dopants into electrically useful lattice positions.

Step 8 Ion Implantation profile : The Implanted dopant normally produce an approximately Gaussian distribution A Simplified expression is:

$$(x) = \frac{Q}{\sqrt{2\pi}\,\Delta R_p} \exp\left[-\frac{(x - R_p)^2}{2(\Delta R_p)^2}\right]$$

where Q = Implanted dose, R_p = projected range
ΔR_p = range Straggle x = depth

Imparted parameters:

Implantation does

$$Q = \frac{\text{number of Implanted ions}}{\text{Unit area}}$$

Usually expressed in ions/cm^2.

Implantation energy

expressed in: KeV or MeV

Higher Implantation energy generally produces a deeper Implant

How ion Implantation Works Inside Silicon.

Flow chart when an energetic ion enters Silicon:

ion enters → Collision with atoms → loses energy → changes direction. → eventually Stops.

There are two major energy-loss mechanism.

Nuclear stopping: The Implanted ion Collides with Silicon nuclei This Causes (i) deflection of the ion (ii) displacement of Silicon atoms (iii) Lattice damages. $S_n = \frac{-1}{N_{Si}} \frac{dE_{ion}}{dx}$

Electronic Stopping: The Implanted ion Interacts with electrons in the Silicon The Causes energy loss through excitation and ionization. At high energies, electrons Stopping Can become particularly Important.

$$S_e = \frac{-1}{N_{Si}} \frac{dE_{ion}}{dx}$$

11

CamScanner

Summary

Diffusion Vs Ion Implantation.

Feature	Diffusion	Ion Implantation
① Dopant Introduction	Thermal movement	Accelerated ion.
② main driving force	Concentration-gradient	Ion Kinetic energy
③ Temperature	high	Implantation can be relatively Low temperature
④ Depth Control	Less precise	Highly precise.
⑤ Dose Control	moderate	Excellent
⑥ Crystal damage	Low	Significant
⑦ Annealing	Sometimes required	Normal required
⑧ profile	erfc / Gaussian	Approximately Gaussian.
⑨ modern IC use	Less dominant for shallow Junction.	Widely used.
⑩ equipment	Diffusion furnace	Ion Implanter

Advantages

① Relatively Simple process ② Can process many wafer Simultaneously ③ Suitable for high-temp processing ④ Good for ~~high~~ Introducing ~~long~~ large dopant Concentration ⑤ Historically Important in Semi-Conductor fabrication

Disadvantages

① Requires high temp ⑪ Dopant spread laterally as well as Vertically ⑪ Less precise depth Control ⑩ Difficult to form extremely shallow junctions ⑤ High-temp exposure can affect previously fabricated Structures.

(12)

CamScanner

Channeling in ion Implantation.

Channeling occurs when ion enter the crystal along certain crystallographic directions and travel relatively deeply through open channels between atoms. This can produce an unexpectedly deep dopant distribution.

To reduce channeling manufacturers may:

1. tilt the wafer relative to the ion beam
2. rotate the wafer
3. use suitable Implantation Conduction.
4. Sometimes introduce a thin amorphous layer.

Example: formation of an n type Region.

Suppose we want to Convert part of a Silicon wafer into an n-type region.

Using diffusion.

Silicon wafer → phosphorus Source → High temp → patom diffuse into Si → n type region.

Using ion Implantation.

phosphorus Source → P^+ ions → mass analysis → Acceleration → Implantation into Si → Annealing → n type region.

$$\therefore C_x = \frac{Q}{\sqrt{2\pi}\Delta R_p} e^{-\frac{(x-R_p)^2}{2(\Delta R_p)^2}}.$$

CamScanner

Practice problem 5:

A Silicon wafer is implanted with a boron dose of $Q = 1.0 \times 10^{13} cm^{-2}$ at an energy where the projected range is $R_p = 0.5 \mu m$ and the straggle is $\Delta R_p = 0.05 \mu m$.

(a) Calculate the peak dopant concentration N_p

(b) Determine the concentration of boron at the surface ($x = 0$)

(a) Calculate peak concentration (N_p)

$$N_p = \frac{Q}{\sqrt{2\pi}\,\Delta R_p}$$

$$N_p = \frac{1.0 \times 10^{13} cm^{-2}}{\sqrt{2\pi} \times (0.05 \times 10^{-4} cm)}$$

$$N_p = \frac{1.0 \times 10^{13}}{2.5066 \times 5.0 \times 10^{-6}} = \frac{1.0 \times 10^{13}}{1.253 \times 10^{-5}} = 7.98 \times 10^{17} cm^{-3}$$

(b) $x = 0$

$$N(0) = N_p \exp\left(-\frac{(0 - R_p)^2}{2\Delta R_p^2}\right) = N_p \exp\left(-\frac{R_p^2}{2\Delta R_p^2}\right)$$

$R_p = 0.5 \mu m$

$\Delta R_p = 0.05 \mu m$

$$\frac{R_p}{\Delta R_p} = \frac{0.5}{0.05} = 10$$

$$\exp\left(-\frac{10^2}{2}\right) = \exp\left(-\frac{100}{2}\right) = 1.93 \times 10^{-22}$$

$$\exp(-50) = 1.93 \times 10^{-22}$$

$$N(0) = (7.98 \times 10^{17}) \times (1.93 \times 10^{-22}) = 1.54 \times 10^{-4} cm^{-3}$$

CamScanner

(2) An n-type silicon substrate with a uniform background donor concentration of $C_B = 1.0 \times 10^{15} cm^{-3}$ is implanted with boron to form a p-type region. The implant results in a peak concentration of $N_p = 1.0 \times 10^{17} cm^{-3}$ at a projected range of $R_p = 0.4 \mu m$ with a straggle of $\Delta R_p = 0.05 \mu m$. Calculate the junction depth (x_j) which is the depth where the implanted acceptor concentration equals the background n-type concentration ($N(x_j) = C_B$).

$$N(x_j) = N_p \exp\left(-\frac{(x_j - R_p)^2}{2\Delta R_p^2}\right) = C_B$$

$$\frac{C_B}{N_p} = \exp\left(-\frac{(x_j - R_p)^2}{2\Delta R_p^2}\right)$$

$$\ln\left(\frac{N_p}{C_B}\right) = \frac{(x_j - R_p)^2}{2\Delta R_p^2}$$

$$x_j = R_p + \Delta R_p \sqrt{2 \ln\left(\frac{N_p}{C_B}\right)}$$

* Ratio of Con. $\frac{N_p}{C_B} = \frac{1.0 \times 10^{17}}{1.0 \times 10^{15}} = 100$
* Natural log term = $\ln(100) = 4.605$
* Multiply by 2 = $2 \times 4.605 = 9.21$
* Square root $\sqrt{9.21} = 3.035$

$$x_j = 0.4 \mu m + (0.05 \mu m \times 3.035)$$
$$= 0.4 + 0.152 = 0.552 \mu m$$

The p-N junction is formed at a depth of $0.552 \mu m$ below the Silicon surface.

CamScanner

(17) you need to Implant a 300mm diameter Silicon wafer with a boron dose of $Q = 2.0 \times 10^{14}$ ions/cm^2.

(a) Calculate the total number of ions required for a single wafer

(b) If the ion Implanter operates with a Singly charged boron ion beam ($q = 1.6 \times 10^{-19}$ C) and you want to process the wafer in 60seconds, what beam Current I must be delivered to the wafer? (Assume a Circular wafer and 100% of the beam hits the active area)

Radium $r = \frac{d}{2} = \frac{300}{2} = 150mm = 15cm$.

Area $= \pi r^2 = \pi \times (15cm)^2 = 225\pi\ cm^2$ or $3.142 \times 225 = 706.86\ cm^2$

Calculate Total number of ions (N_{total})

$N_{total} = Dose(Q) \times Area(A)$

$= (2.0 \times 10^{14}\ ions/cm^2) \times 706.86\ cm^2$

$N_{total} = 1.414 \times 10^{17}$ ions

(b) Cal the beam $I = \frac{Q_{total}}{t}$

Q total (charge) $= N_{total} \times q = 1.414 \times 10^{17} \times 1.6 \times 10^{-19} C$

$Q_{total} = 0.0226$ Coulombs

$I = \frac{0.0226}{60s} = 0.000377\ A = 0.377\ mA$

Note
The Implanter requires a beam Current of 0.377 mA or 377μA to achieve the target dose in one minute.

CamScanner

During an Ion Implantation step; you need to protect certain area of a Silicon wafer using a photoresist mask. The wafer is Implanted with boron ions at a high energy where the projected range in photoresist is $R_p; P_R = 0.8\mu m$ and the straggle is $\Delta R_{p P_R} = 0.08\mu m$. If the underlying substrate must not receive more than 0.1% of the peak Implant dose what is the minimum required thickness of the photoresist mask (x_{mask})?

$$\frac{N(x_{mask})}{N_p} = \exp\left(\frac{-(x_{mask} - R_p P_R)^2}{2\Delta R^2_{p P_R}}\right)$$

Rearrange to solve for mask thickness

$$\frac{-(x_{mask} - R_p P_R)^2}{2\Delta R^2_{p P_R}} = \ln 10^{-3}$$

Since $\ln 10^{-3} = 3\ln(10) \simeq -6.908$ the negative signs cancel out

$$\frac{(x_{mask} - R_p P_R)^2}{2\Delta R^2_{p P_R}} = 6.908$$

~~$\therefore 2\Delta R^2_{p P_R} = 2$ both Square Root~~ multiple by $2\Delta R^2_{p P_R}$ and take the Square Root.

$$x_{mask} - R_p P_R = \sqrt{2 \times 6.908} \times \Delta R_{p P_R}$$

$$x_{mask} = R_p P_R + \Delta R_{p P_R} \sqrt{13.816}$$

$\therefore \sqrt{13.816} = 3.717$

$= x_{mask} = 0.8\mu m + (0.08\mu m * 3.717$

$= 0.8 + 0.297 = 1.097\mu m$

To Safely block the boron ion and keep transmission under 0.1% the photoresist mask must be at least 1.10 μm Thick In real-world manufacturing.

Engineer minimum aiming 1.3 or 1.4μm: